

RESEARCH ARTICLE

The wettability of pure liquid aluminum on oxide substrates

Xingwen Wei¹  | Beate Fankhänel² | Wei Zhao³ | Anton Perminov¹ |
 Enrico Storti⁴ | Mengke Liu⁵  | Guojun Ma⁵ | Olga Fabrichnaya⁶ |
 Alexandros Charitos² | Olena Volkova¹

¹Institute of Iron and Steel Technology, Technical University Bergakademie Freiberg, Freiberg, Germany

²Institute of Nonferrous Metallurgy and Purest Materials, Technical University Bergakademie Freiberg, Freiberg, Germany

³State Key Laboratory of Bioelectronics, School of Biological Science & Medical Engineering, Southeast University, Nanjing, China

⁴Institute of Ceramics, Refractories and Composite Materials, Technical University Bergakademie Freiberg, Freiberg, Germany

⁵Joint International Research Laboratory of Refractories and Metallurgy, Ministry of Education, Wuhan University of Science and Technology, Wuhan, Hubei, China

⁶Institute of Material Science, Technical University Bergakademie Freiberg, Freiberg, Germany

Correspondence

Xingwen Wei, Institute of Iron and Steel Technology, Technical University Bergakademie Freiberg, Leipziger Str. 34, Freiberg 09599, Germany.
 Email: xingwen.wei@extern.tu-freiberg.de

Funding information

German Research Foundation (DFG); Technical University Bergakademie Freiberg

Abstract

Oxide particles are considered to be one of the reasons for the stability of cellular porous Al foam. Thus, the contact angle (θ) of molten pure aluminum on oxide substrates, including Al_2O_3 , $\text{CaO} \cdot 2\text{Al}_2\text{O}_3$ (CA2), $\text{CaO} \cdot 6\text{Al}_2\text{O}_3$ (CA6), mullite ($3\text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2$), and SiO_2 was measured in a vacuum heating microscope at a temperature of 950°C to determine the value of wettability of molten aluminum on various oxides. Thereafter, the measured contact angles followed the tendency as mullite (155°) > CA6 (152°) > MgAl_2O_4 (151°) > Al_2O_3 (149°) > SiO_2 (124°) > CA2 (100°). Furthermore, the cross-sectional area between molten aluminum and substrates was thoroughly analyzed, and the reaction products, such as the Al_2O_3 layer at the interface, have been detected. Thermo-Calc, with the combination of available databases, was used to perform thermodynamic simulations and to discuss and predict possible reactions under the current experimental conditions.

KEYWORDS

Al foam, Al_2O_3 layer, contact angle, stability of foam

1 | INTRODUCTION

Aluminum foam materials have found wide application in the aerospace, automobile, and construction industries due to their excellent thermal resistance, mechanical

properties, and lightweight nature.^{1–4} Various production methods have been reported over the last few decades, including the addition of a blowing agent^{3,5–7} and gas injection.^{2,8,9} Especially, the gas injection method has unique advantages, including low cost and the ability to

This is an open access article under the terms of the [Creative Commons Attribution-NonCommercial-NoDerivs](https://creativecommons.org/licenses/by-nc-nd/4.0/) License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.

© 2024 The Author(s). *Journal of the American Ceramic Society* published by Wiley Periodicals LLC on behalf of American Ceramic Society.

control parameters such as cell size of the Al foam.¹⁰ To produce metal foams using the gas injection method, stable foams can only be achieved with the presence of solid particles,^{2,11,12} since the solid particles enhance viscosity^{13,14} and foam stability.¹⁵

Solid particles, including SiO_2 , SiC , Al_2O_3 , MgO , and MgAl_2O_4 , were used to stabilize the Al alloy foam.^{10,11,15–17} Interior solid particles are frequently detected in the foam, and their composition is related to the composition of the alloy. Kadoi and Nakae¹⁵ investigated the effect of various oxide particles (Al_2O_3 , SiO_2 , and MgO) on the stability of Al foam. Kumar et al.^{17,18} studied the impact of MgAl_2O_4 particles on the stability of Al-11 wt.%Si foams in various concentrations. Besides, Bhosale et al.¹⁹ used ultrafine MgAl_2O_4 particle to increase the stabilization of Mg–3Ca alloy foams. Heim et al.²⁰ prepared foams by injecting gas and adding particles in combination.

According to the theory proposed by Minto and Davenport and discussed in the work of Ip et al.,¹¹ a stable bubble-particle attachment could occur if the solid particle is not wetted by the liquid. However, the question arises as to why some particles are more efficient at stabilizing the Al foam. In the previous work by Haibel et al.,²¹ many scenarios were discussed to reveal the stabilization mechanism. He stated that when the wettability of the liquid Al on the oxides is around 90° (the boundary between wetting and nonwetting), the oxides are still well embedded on the bubbles surface between the bubbles. If the liquid Al has a high wettability on the oxides, the particles can form a large meniscus on the bubble surface, leading to bubble rupture. Kaptay¹⁶ assumed a suitable contact angle range for foam stabilization is 50° – 90° according to his calculation.

The calcium oxide (CaO) and calcium aluminates have frequently been observed when Ca is present in the alloys. Calcium has the function of increasing the viscosity of aluminum melts, and low surface tension causes a reduction in the critical cell wall thickness when collapse occurs. Moreover, the addition of calcium reduced the contact angle of molten aluminum. In the study of Ip et al.,²² it was found that the addition of 1–2 wt.% calcium reduced the contact angle on the Al_2O_3 substrate at 900°C from 100° – 110° to 90° – 95° . The contact angle between molten aluminum and CaO showed an average value of 100° . The significance of wettability of molten Al alloys on various oxides is crucial for the tailoring of Al foam.

Sessile drop method is known as a method that can contribute to the measurement of surface tension, contact angle, and density. In addition, the sessile drop method is also used to investigate corrosion and joining mechanism.²³ It is widely used due to its simple design and experimental accuracy. In the conventional sessile drop method, the gas atmosphere is one of the most influential factors that could affect the results, including the wetting angles, surface tension, density measurements, as well

as the interactions. For instance, the interaction process between Fe and Al_2O_3 could strongly affect the wettability of molten iron on the substrate due to the formation of the compound hercynite (FeAl_2O_4). Therefore, to avoid this, the oxygen partial pressure should low enough to avoid the formation of hercynite. Argon, helium, nitrogen, and so forth are commonly used as inert gases to provide a nonoxidizing atmosphere, but technically a 100% pure inert gas does not exist. A gas purifier (with Mg, Ti sponge) is introduced in some sessile drop experiments to achieve a more satisfying atmosphere. These additional requirements are usually applied to nonferrous metals such as Mg and Al, which oxidize even at room temperature. Bao et al.²⁴ investigated the wettability of Al on alumina under an argon atmosphere combined with a gas purifier and under a vacuum atmosphere. Finally, due to the oxide formation on the surface of the molten aluminum, the author selected the vacuum atmosphere for further wetting investigations. However, Bao et al. found out that the vacuum atmosphere is also problematic due to possible evaporation in the liquid state and the low-pressure conditions. Besides, as pointed out by Sobczak et al.²⁵ when graphite heating elements are used, the presence of a thin carbon layer could considerably enhance the wetting between Al and Al_2O_3 materials. The effect of carbon on the wetting of aluminum oxides by Al was studied in detailed by Weirauch and Krafick.²⁶

To deeply understand the relevance between the particles and the stability of the Al foams, good control of bubble injection through the orifice,²⁷ as well as guidance of the particle selection to stabilize the foam materials, is significant. The wetting behavior of molten aluminum on different ceramic materials is extremely important. However, due to the high oxidation nature of the aluminum-based alloys, only a limited number of literatures related to its wettability on various oxides have been reported. In the present study, pure aluminum was investigated on various ceramic substrates including Al_2O_3 , $\text{CaO}\cdot 2\text{Al}_2\text{O}_3$ (CA2), $\text{CaO}\cdot 6\text{Al}_2\text{O}_3$ (CA6), mullite, SiO_2 , and MgAl_2O_4 . The choice of ceramic materials is based on the possible formation of oxides in the liquid Al-based alloy. The main difficulty of the present investigation is the strong affinity of the present pure aluminum for oxygen. A vacuum sessile drop method was used for the experiments to avoid the unwanted oxidation, as shown in the experimental section.

2 | EXPERIMENTAL PROCEDURE

The raw materials used to prepare the ceramic substrates are listed in Table 1. The additives were dextrin (used as a binder for the granulation), Optapix AC95 (for dry pressing) and PAF35 (Zschimmer & Schwartz—used as temporary binders), and Zusoplast G112 (a pressing and lubricating agent for use in ceramic auxiliary applica-

TABLE 1 Physical properties of the various ceramic substrates.

Raw materials	Mass phase fractions (wt.%) by XRD	Roughness (Sa) (μm)	Surface roughness parameter Sdr (%)	Open porosity
Aluminum oxide	Alpha-alumina 100	3.23	29	24.3
Calcium hexaluminate CA6	Calcium hexaluminate 95, alpha-alumina 5.0	1.57	19	26.9
Calcium dialuminate CA2	CA2 100	1.87	7.6	<1
Mullite	Mullite 91.5, alumina 8.5	2.85	33.65	28.8
Silica	Cristobalite 100	12.28	201.5	24.6
MgAl ₂ O ₄	Spinel, 100	2.04	47.58	18.9
Raw materials used for the silica substrates				
Material	Denomination		Amount	
Fused silica (fine)	Tecosil 021		30%	
Fused silica (coarse)	Tecosil 017		65%	
Microsilica (filler)	RW		5%	
Silica sol (liquid binder)	Köstrosol 1470		7% (rel. to solids)	

tions). In general, the granulates were prepared by mixing the raw materials with 2 wt.% dextrin. And 2 wt.% of deionized water was slowly added by spraying. Afterward the compaction behavior of the granules was tested by hand-pressing. Ceramic substrates with dimensions of 50 mm (diameter) $\times$ 8 mm (height) were pressed uniaxially with an applied pressure of 120 MPa. After drying at 120°C, sintering was performed at 1600°C in an electric furnace at a heating rate of 1 K/min and a dwell time of 5 h. Since CA2 hydrates slowly in the presence of water, only ethylene glycol (2 wt.%) was used instead of dextrin + water. As ethylene glycol is quite sticky, no additional binder was required. About 33 g of mixture was used for each sample. Sintering was carried out as described above. Mixing for the silica substrates was done starting from the coarse fraction. Silica sol was used to granulate the powders for pressing. Pressing, drying, and sintering were carried out as described above.

The surface of the tablets was examined by confocal laser scanning microscopy using a confocal microscope VK-X1000 from Keyence, Japan at 20 $\times$ magnification. Three samples per batch were analyzed in total. For each polished sample, an area of 5 $\times$ 5 mm² was scanned, and a 3D profile was reconstructed from a total of 99 adjacent sectors. The final 3D profile was analyzed by means of the MultiFileAnalyzer software, as shown in Figure 1. The open porosity and bulk density of the samples were estimated by water immersion according to the European standard DIN EN 993-1.²⁸ Further details have been reported in the previous work on the preparation of pure Al₂O₃.^{28–30} The final substrate phases were analyzed by X-ray diffraction (XRD). The instrument (X'Pert PRO MPD X-Ray Diffractometer 3040/60, PANalytical) operates in Bragg–Brentano geometry with a fixed divergence slit and a rotating sample stage. The oxides measured by XRD are presented in Figure 2.

2.1 | Sessile drop setup

The heating microscope for the sessile drop method consists of a high temperature furnace (MoSi₂ as a heating element), and a high-vacuum system from Gero GmbH was used to perform the wetting experiment by contact angle measurements as shown in Figure 3. Detailed information regarding the construction of the sessile drop apparatus has been reported in the literature.^{31,32} Pure aluminum (99.99% purity) was used for the present investigation under a vacuum atmosphere ($p \leq 1.5 \times 10^{-3}$ Pa). For each experiment, an aluminum sample (approximately 100 mg) was placed on top of the ceramic substrate in the furnace. The metal samples were cut on all of their faces and mechanically polished immediately before being placed in the furnace to obtain fresh surfaces with minimal oxide coverage.

For the present work, the selected temperature is the closest value to the Al foam preparation that the presence furnace allows (950°C). When using the classical sessile drop technique to measure contact angles with aluminum, a high temperature and a high vacuum pressure are required to remove the oxide layer on top of the aluminum melt and to avoid reoxidation of the aluminum droplet during the experiments. Since the highest vacuum pressure in the furnace that can be achieved in a reasonable time is $p \leq 1.5 \times 10^{-3}$ Pa, a temperature of $T \geq 950^\circ\text{C}$ is required to reduce the Al₂O₃ to the gaseous Al₂O. A detailed description of this mechanism can be found in Fankhänel et al.³¹ The microscope was heated up with a heating rate of 350 K/h up to target experiment temperature of 950°C and held for 90 min. Moreover, the interior pressure of the experimental chamber was continuously monitored by the pressure detector. For the sake of reliability of the results, each substrate sample was repeated

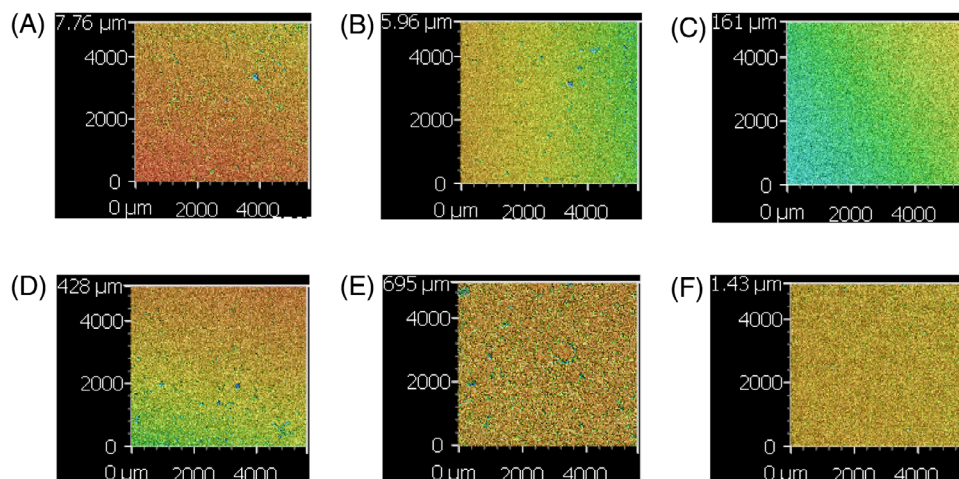


FIGURE 1 3D profiles collected from pure substrates: (A) Al_2O_3 , (B) CA6, (C) CA2, (D) mullite ($3\text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2$), (E) SiO_2 , and (F) MgAl_2O_4 . The colors indicate the depth, where the transition from red to green is smooth; it means that the surface is not exactly flat. Spot indicates the pore in the sample.

up to three times, an average result was then presented below.

After the wetting experiments, the samples (solidified aluminum droplets on oxidic substrates) were integrally embedded in the epoxy resin and then cut perpendicularly and polished for the cross-sectional analysis. Scanning electron microscopy in combination with energy-dispersive X-ray spectroscopy was used to determine the morphology and chemical composition (Ultra55, Zeiss NTS GmbH).

2.2 | Thermodynamic simulation

Thermodynamic databases of the Mg–Al–O, Ca–Al–O, and Al–Si–O systems were derived by combining binary descriptions with quasi-binary oxide systems. The thermodynamic descriptions were accepted for the MgO– Al_2O_3 system were taken from ref. [33], Mg–Al from ref. [34], Al–O from ref. [35], and Mg–O from ref. [36], CaO– Al_2O_3 ,³⁷ Ca–O,³⁸ Al_2O_3 – SiO_2 ,³⁹ and Al–Si.⁴⁰ Gas phase thermodynamic data for the gas phase was accepted from SSUB SGTE database.⁴¹ All including binary and quasi-binary systems were checked to reproduce the phase diagrams of the corresponding systems before simulating the interaction between molten aluminum and ceramic substrates. The calculations for corundum Al_2O_3 –Al, CA2–Al, CA6–Al, mullite–Al, SiO_2 –Al, spinel MgAl_2O_4 –Al were performed at 950°C and pressure variation in the range from 1×10^{-4} to 1 Pa. For spinel, phase calculations were performed at higher pressures up to 100 Pa to determine at which pressures gas phase appears. The spinel substrate contained excess of Al_2O_3 , that is, 58.36 mol% in compar-

ison with stoichiometric spinel (50 mol%). At condition of synthesis, Al_2O_3 enriched spinel is a single phase, but at 950°C the excess Al_2O_3 should be precipitated forming two-phase mixture $\text{MgAl}_2\text{O}_4 + \text{Al}_2\text{O}_3$. At low pressure, spinel is not stable. The MgO content in spinel phase decomposes forming Mg rich gas while the released oxygen reacts with liquid Al and thus increases Al_2O_3 fraction due to oxidation of Al. In the Al–O system, the gas, which appears in equilibrium with liquid Al and Al_2O_3 , is enriched by Al 92 mol% and contains 8 mol% of Al_2O .

According to the calculations, obvious reactions between substrate and Al occur for mullite and silica at pressure of 0.1 Pa. In the case of mullite and silica, the SiO_2 is reduced to metallic Si by Al melt. In the case of SiO_2 and mullite, the products formed depend on the ratio of Al to substrate. In all calculations, the ratio of substrate to Al was taken 1.5:1. While, for the case of SiO_2 and mullite substrates, the influence of the substrate/Al ratio is demonstrated. Thermodynamic modeling of mullite reproduces experimentally observed narrow homogeneity range varying with temperature, but the composition of the sample corresponds to stoichiometric mullite $\text{Al}_6\text{Si}_2\text{O}_{13}$, which should be stable under sintering condition at 1600°C . However, according to XRD, the sample consists of mullite and Al_2O_3 , which probably precipitated during cooling. In the case of CA2 and CA6, calcium aluminates react with Al to a very small extent, dissolving Ca in liquid Al in small amounts (0.23 wt.%) and forming small fractions of CA6 and Al_2O_3 , respectively. At $p = 5 \times 10^{-3}$ Pa, CA2 reacts with liquid Al, forming gas and a CA6 phase which enriched in Al (65.95 mol%), but also contains Ca (29.26 mol%) and Al_2O (4.88 mol%). At the same pressure, CA6 reacts with the

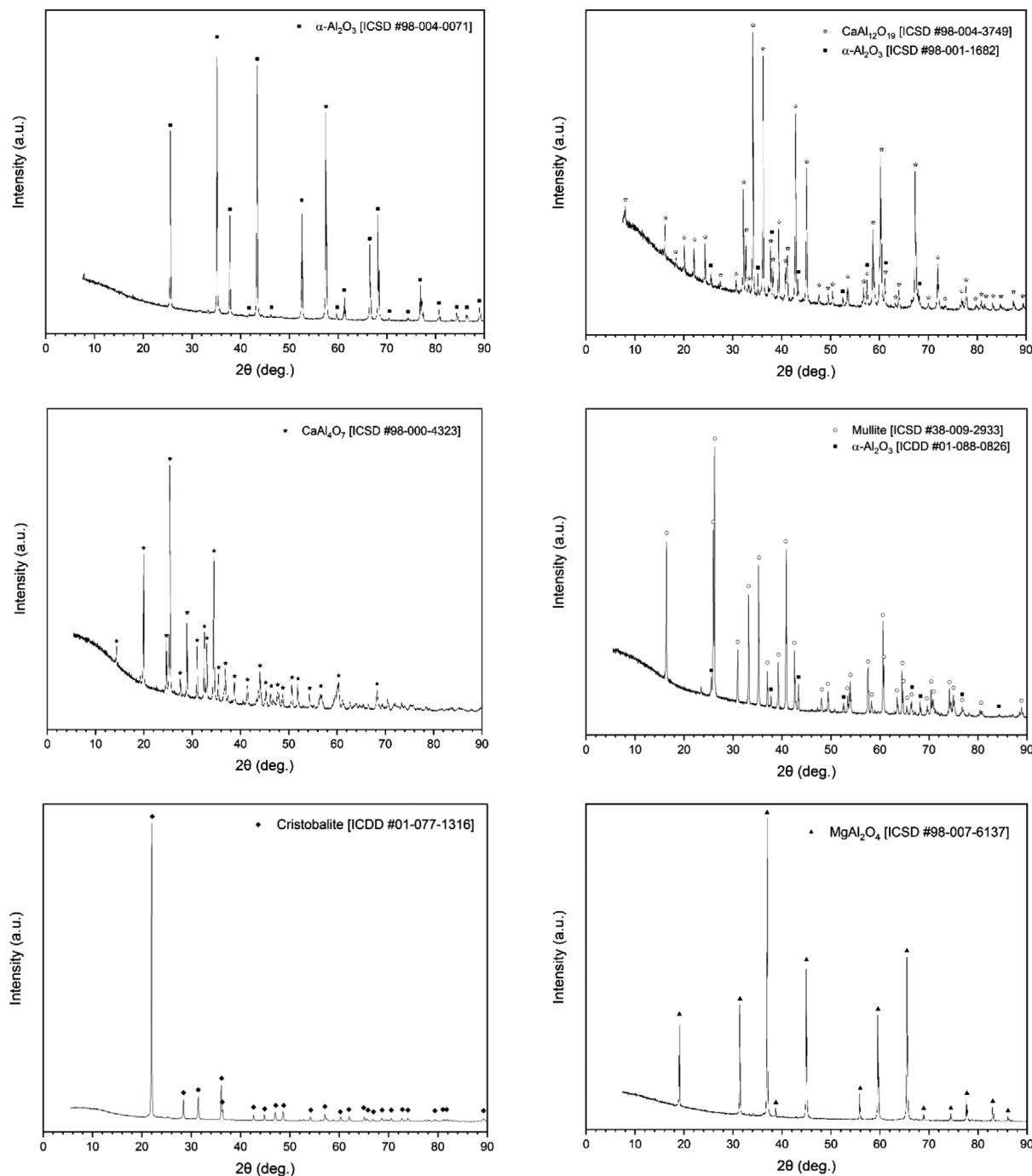


FIGURE 2 X-ray diffraction profiles collected from pure substrates.

liquid forming gas (79.54 mol% Al, 13.89 mol% Ca, and 6.57 mol% Al_2O_3) and increases the fraction of corundum. The results of the equilibrium phase composition calculations are presented in Figure 4 and, for detailed information, in Table 2. The pressure for the calculation was chosen to be higher and lower than that one at which the gas formed. The precise composition of the solid phase and the gas phase is presented at selected pressures in Table 2.

3 | RESULTS AND DISCUSSION

Figure 5 presents the molten Al droplets on various ceramic substrates. For the ceramic substrates such as Al_2O_3 , CA6, mullite, and MgAl_2O_4 , the contact angle of the liquid aluminum showed a significantly high value. On the contrary, the liquid aluminum spread the most on the CA2 substrate and showed a slightly higher wettability than on the SiO_2 substrate. Furthermore, the mullite substrate even

TABLE 2 Calculated equilibria between ceramic and molten aluminum with various substrate/molten Al ratio under different pressures at 950°C.

Substrate	Ratio substrate/Al	P, Pa	Phase and fraction (%)
Al ₂ O ₃	1.5:1	1 × 10 ⁻²	Al ₂ O ₃ 60 Al 40
		1 × 10 ⁻³	Al ₂ O ₃ 59.36 Gas 40.64 (Al 95.12 mol%, Al ₂ O 4.88 mol%)
CA2	1.5:1	1 × 10 ⁻²	CA2 59.79 Al 40.03 (Al 99.89 mol%, Ca 0.11 mol%,) CA6 0.18
		1 × 10 ⁻³	CA2 51 Gas 49 (Al 67.93 mol%, Ca 29.01 mol%, Al ₂ O 3.06 mol%)
CA6	1.5:1	1 × 10 ⁻²	CA6 44.56 Al ₂ O ₃ 15.44 Al 40 (Al 99.98 mol%, Ca 0.02 mol%)
		1 × 10 ⁻³	Al ₂ O ₃ 55.21 Gas 44.79 (Al 80.75 mol%, Ca 15.33 mol%, Al ₂ O 3.92 mol%)
Mullite	1.5:1	1 × 10 ⁻¹	Mullite 35.60 (Al ₂ O ₃ 56.17 mol%, SiO ₂ 43.83 mol%) SiO ₂ 14.40 Si 50
		1 × 10 ⁻²	Al ₂ O ₃ 20 Gas 80 (SiO 100 mol%)
	1.0:1.0	1 × 10 ⁻¹	Mullite 29.26 (Al ₂ O ₃ 57.28 mol%, SiO ₂ 42.72 mol%) Si 62.5 Al ₂ O ₃ 8.24
		1 × 10 ⁻²	Al ₂ O ₃ 25 Si 41.66 Gas 33.34 (SiO 99.99 mol%, Al 0.006 mol%, Si 0.004 mol%)
	1:1.5	1 × 10 ⁻²	Al ₂ O ₃ 26.67
		1 × 10 ⁻³	Si 59.35 Liquid 13.98 (Al 60.28 mol%, Si 39.72 mol%) Al ₂ O ₃ 26.37 Si 65.75 Gas 7.88 (Al 88.20 mol%, SiO 7.36 mol%, Al ₂ O 4.41 mol%, Si 0.03 mol%)
SiO ₂	1.5:1	1 × 10 ⁻²	Al ₂ O ₃ 54.24 Si 10.20 Liquid 35.56 (Al 60.28 mol%, Si 39.72 mol%)
		1 × 10 ⁻³	Al ₂ O ₃ 53.49 Si 26.47 Gas 20.03 (Al 88.20 mol%, SiO 7.36 mol%, Al ₂ O 4.41 mol%, Si mol%)
	2:1	0.1	Al ₂ O ₃ 51.92 Si 41.67 Mullite 6.41 (Al ₂ O ₃ 57.28 mol%, SiO ₂ 42.72 mol%)
		0.01	Al ₂ O ₃ 55.60 Si 37.1 Gas 7.3 (SiO 99.99 mol%, Al 0.006 mol%, Si 0.004 mol%)
Spinel	1.5:1	100	Spinel 50.07 (Al ₂ O ₃ 50.205 mol%, MgO 49.795 mol%) Al ₂ O ₃ 9.9 Liquid 40.03 (Al 99.87 mol%, Mg 0.13 mol%)
		10	Al ₂ O ₃ 44.34 Liquid 23.35 (Al 99.95 mol%, Mg 0.05 mol%) Gas 33.31 (Mg 99.95 mol%, Al 0.046 mol%, Al ₂ O 0.004 mol%)

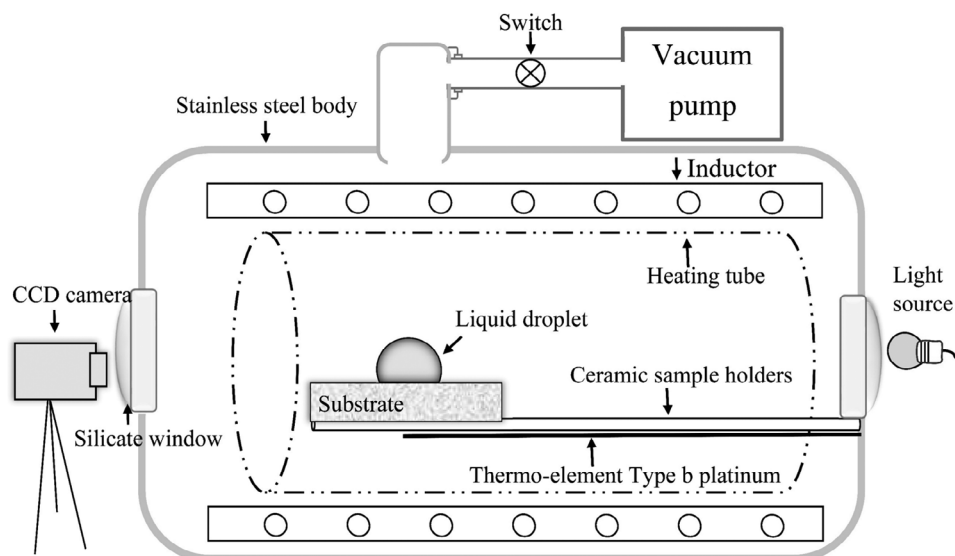
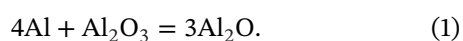


FIGURE 3 The schematic illustration of the vacuum sessile drop setup.

exhibited a reactive phenomenon. Some reaction products were observed on the liquid aluminum droplet and on the surface of the mullite.

3.1 | Monitored pressure

The pressures measured during the experiments are plotted in Figure 6. It can be seen that in all cases, the pressures decreased to approximately 10^{-5} mbar at room temperature before heating. As the temperature increases, the pressures also increased. In this case, the pressures exhibited different values. The measured pressures followed a sequence as $\text{Al}_2\text{O}_3 > \text{CA6} > \text{MgAl}_2\text{O}_4 > \text{Mullite} > \text{SiO}_2 > \text{CA2}$. Since the formation of Al_2O (via Equation (1)) gas is only possible when the temperature reaches approx. 950°C under the present pressure, the organic additives in the substrate sample are considered to be the main reason for the higher pressure at lower temperature.³¹ Similar results were observed in the work of ref. [31]. Furthermore, the measured pressures at the experimental temperature showed the different views. Their values could be attributed to the factors such as the porosity of the substrates, the formation of the volatile reaction product between molten Al and substrates, since at the low experimental pressure, the oxides could decompose and further increase the pressure in the chamber, and also the compactness of the substrates may have an essential impact on the pressure.



3.2 | Contact angle

As stated by Shen et al.,⁴² the wetting process (for instance, SiO_2 and MgO systems) could have two different stages, namely, the transient constant stage and the long reaction stage with a decreasing contact angle. While, the intrinsic contact angle is the value that is located in the transient stage. Hence, the selection of the contact angle was based on the consideration of avoiding the effect of the reaction between molten aluminum and the dissolution of the oxide to molten aluminum on the value of the contact angle. Consequently, the contact angle value of the molten aluminum on the substrates was taken from the primary time point, when the oxide layer had completely disappeared and a round droplet had formed. In this results section, the contact angles measured here are comprehensively compared with the reported contact angle of molten aluminum or aluminum-based alloys (due to the lack of information for pure aluminum on some substrates).

Figure 7 presents the measured contact angle of molten Al on different substrates, all of which show a non-wetting behavior (all contact angles are higher than 90°). The contact angles on the Al_2O_3 , CA6, mullite, and MgAl_2O_4 substrates exhibit values around 150° , and the most significant value shown on the mullite substrate is 155° . On the other hand, the contact angle on the CA2 substrate exhibited the lowest value of 100° . The contact angle on the SiO_2 substrate had a value of 124° . The small contact angle on the silica substrate can be predicted by the strong reaction and large dissolution of Si in molten Al (see Table 2). The silica substrate in the previous comparative studies has a smaller roughness than the roughness of the silica in the present

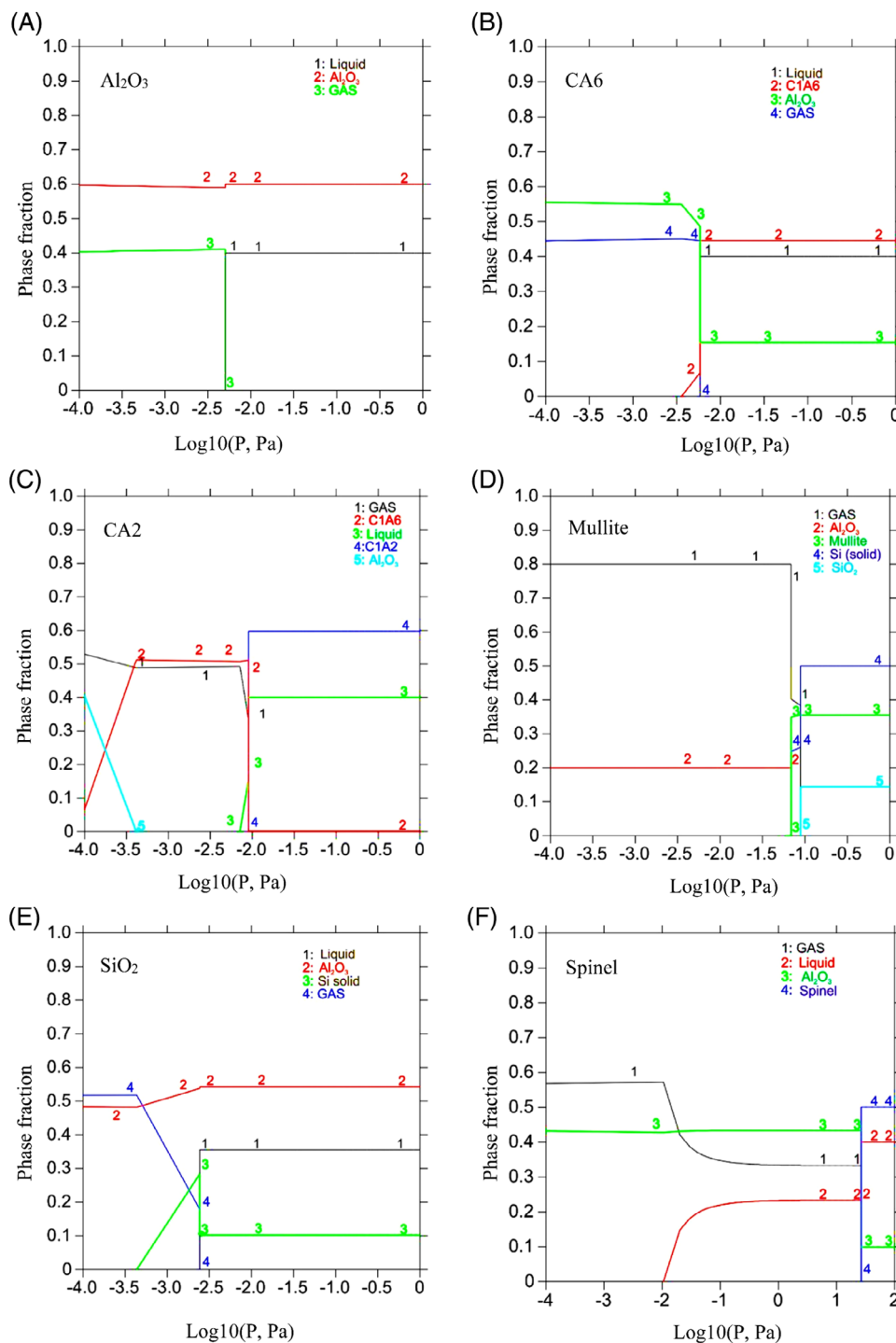


FIGURE 4 Calculation for interaction between molten Al and substrates: (A) Al_2O_3 , (B) CA6, (C) CA2, (D) mullite, (E) SiO_2 , and (F) MgAl_2O_4 .

study. The Wenzel model describes that the droplet is in uniform contact with the rough surface.⁴³ According to Wenzel's equation as shown in Equation (1), the roughness of the substrate has an effect on the contact angle value. To eliminate the roughness effect, Equation (2) was used to calculate the equilibrium contact angle. The roughness r is determined by using the measured S_{dr} , as shown in

Equation (2). The corrected contact angles show reduced values and were summarized with the measured contact angles (see Figure 7). In particular, the corrected contact angle determined for the SiO_2 substrate showed a value of 100.7° close to the value on the CA2 substrate of 99.3° .

$$\cos \theta_w = r \cos \theta_y, \quad (2)$$

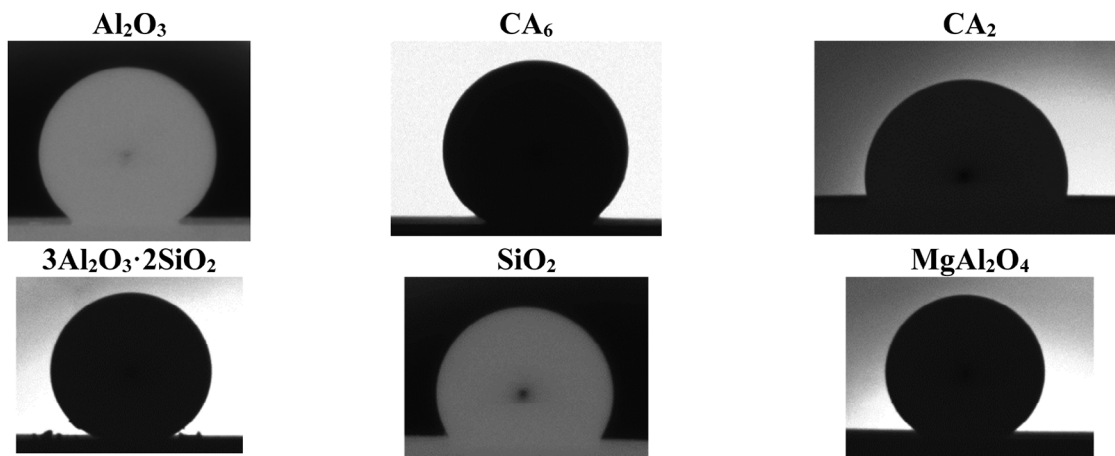


FIGURE 5 Images of the Al droplets on various ceramic substrates.

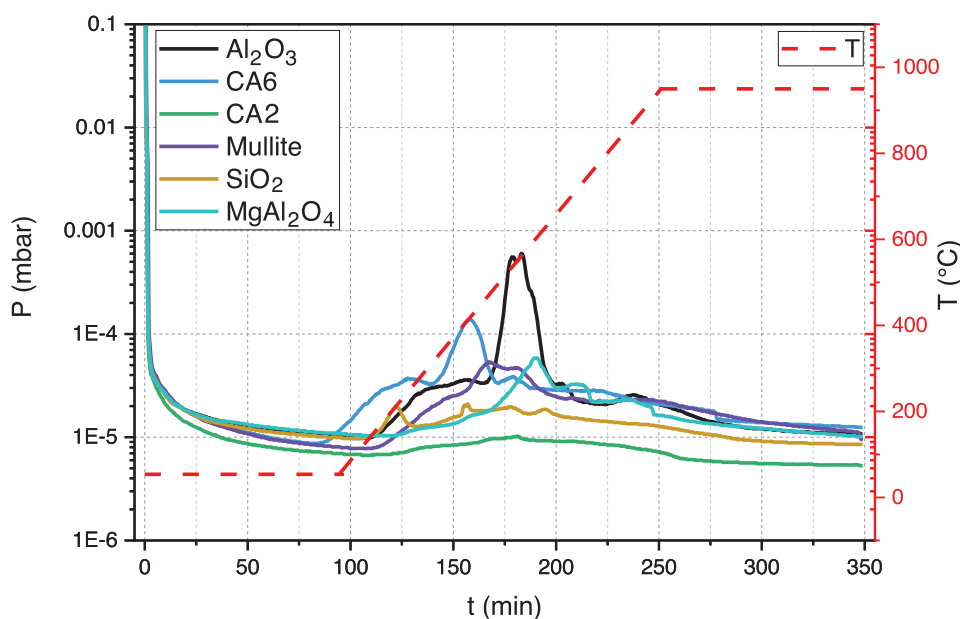


FIGURE 6 The monitored pressures during the experiments with various ceramic substrates.

$$r = \frac{S_{dr}}{100} + 1, \quad (3)$$

where θ_w is the contact angle ($^\circ$) on the rough surface; θ_y is the equilibrium contact angle ($^\circ$); and r is the roughness (μm).

Nevertheless, the present contact angles of molten pure aluminum on the investigated substrates show much higher values than the most values reported in literatures at 950°C . The high contact angle values have been attributed to the oxide layer on the Al droplet at the present temperature. Pure Al is a metal with an extreme affinity for oxygen; the formation of an oxide layer affects the wetting behavior and is very common in such experiments.

The measured and corrected contact angle values are not real or, more accurately, do not show the exact wettability of molten Al on these substrates. Similar results have also been reported in many previous studies at low experimental temperatures, especially below 1000°C . The CP method (capillary purification with no contact) allows the wetting experiments of molten Al down to the melting point, as reported by Voigt et al.⁴⁴ In this case, the measured contact angle of molten Al7SiMg on different substrates kept almost constant up to 30 min. The interaction between molten Al and substrates was slightly observed. These results differ from the similar interaction experiments at higher temperature and the conducted thermodynamic simulations. The reason for this is quite simple; at this

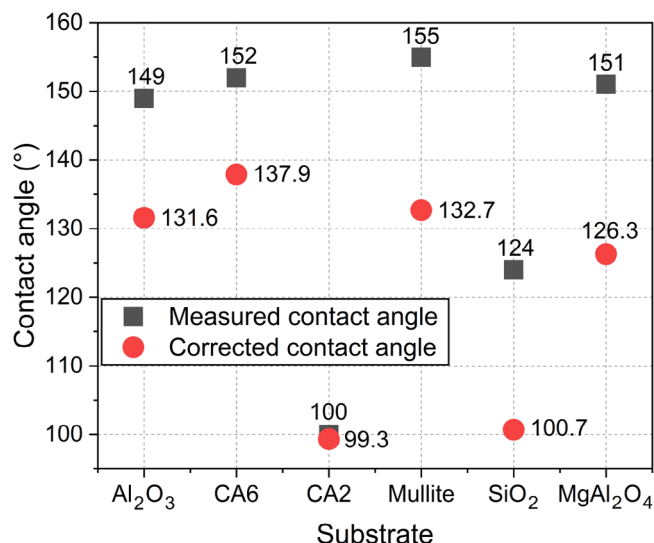


FIGURE 7 The contact angles of molten aluminum on various substrates at 950°C.

low temperature, the oxygen contained in the protective atmosphere reacted with the molten Al as it was pressed out of the injector. The previous contact angle was hardly affected by the oxide layer due to the short contact time between the atmosphere and the droplet. However, with the longer time, the oxide layer was strengthened and consequently prevented a direct contact between the molten Al and the substrate. The present measured contact angles were affected by the oxide layer, whereas the determined contact angle values are still representative and could be used to select stable oxide agents.

3.2.1 | Contact angle on Al₂O₃ substrate

Figure 8 summarizes the measured contact angle of liquid aluminum on alumina and the values from the literature at different temperatures.^{24,26,45–48} The values revealed a decreasing tendency on alumina oxides as a function of temperature. The decreasing tendency of the contact angle has also been stated by Bao et al.²⁴ after reviewing and summarizing the wettability of molten Al on alumina. In addition, John and Hausner⁴⁵ and Laurent et al.⁴⁸ reported a significant reduction in contact angle under a high-vacuum pressure. While John and Hausner⁴⁵ measured a reduction from approx. 125° to approx. 78° under an ultrahigh vacuum atmosphere (reported as 1 μPa) at 700°C, Laurent et al.⁴⁸ measured a contact angle of approx. 88° at 950°C and $p = 2\text{--}5 \times 10^{-5}$ Pa. Both groups stated that the decreased contact angle is resulted from the temperature-induced reaction between molten aluminum and alumina oxides, that is, the formation of suboxide and the evaporation of the molten aluminum.²⁴ With

higher temperature, these processes would be enhanced, which result in a decreased tendency of the contact angle. Therefore, the contact angle of melts on substrates could be varied with the measurement conditions.

To the best of the present author's knowledge, no available contact angle value was found for CA2 and CA6. Consequently, it is not possible to compare the current value with a reported value. A study⁴⁹ of the reaction between molten aluminum and CaO has been reported and will be discussed below.

3.2.2 | Contact angle on mullite and SiO₂ substrate

The interaction of molten aluminum with mullite and SiO₂ substrates has been intensively studied.^{31,42,44,50–53} Shen et al.⁴² interacted molten Al and SiO₂ at different temperatures and summarized the reported wetting values between Al and SiO₂ substrates. These studies showed highly scattered contact angle values. The different results could be attributed to factors such as experimental temperature, atmosphere, substrate properties (i.e., porosity, roughness, and the preparation method⁵¹), and even the measurement technique.⁵⁴ The contact heating (CH) and CP methods were reported to show very different contact angle values between Al and SiO₂. At 800°C, the CH method revealed a contact angle value of 112°, whereas CP method showed a contact angle value of 57°. Table 3 summarizes the reported contact angle of molten Al on mullite and SiO₂ substrates. The collected contact angle between molten Al and mullite showed highly scattered values. However, Marumo and Pask,⁵⁶ Shen et al.⁴² showed the same and close contact angle to the present work in the case of Al/SiO₂ at similar experimental temperature and holding time.

3.2.3 | Contact angle on MgAl₂O₄ substrate

Shinozaki et al.⁵⁹ measured the contact angle of pure aluminum on Al₂O₃-(10, 28 mass pct) MgO substrates at different temperatures from 750 to 1000°C. Furthermore, the interaction of an AlSi7Mg alloy on MgAl₂O₄ has also been studied and showed a similar contact angle at similar temperatures.^{31,44,32} Both Shinozaki et al. (Al₂O₃-(10 mass pct) MgO)⁵⁹ and Fankhänel et al.³¹ reported the reduced contact angle with longer experimental time. For Shinozaki et al. (Al₂O₃-(28 mass pct) MgO),⁵⁹ the contact angle was kept constant at 150°, which is very close to the presently measured value and the value from the study of Fankhänel et al.³¹ with AlSi7Mg at the identical temperature of 950°C.

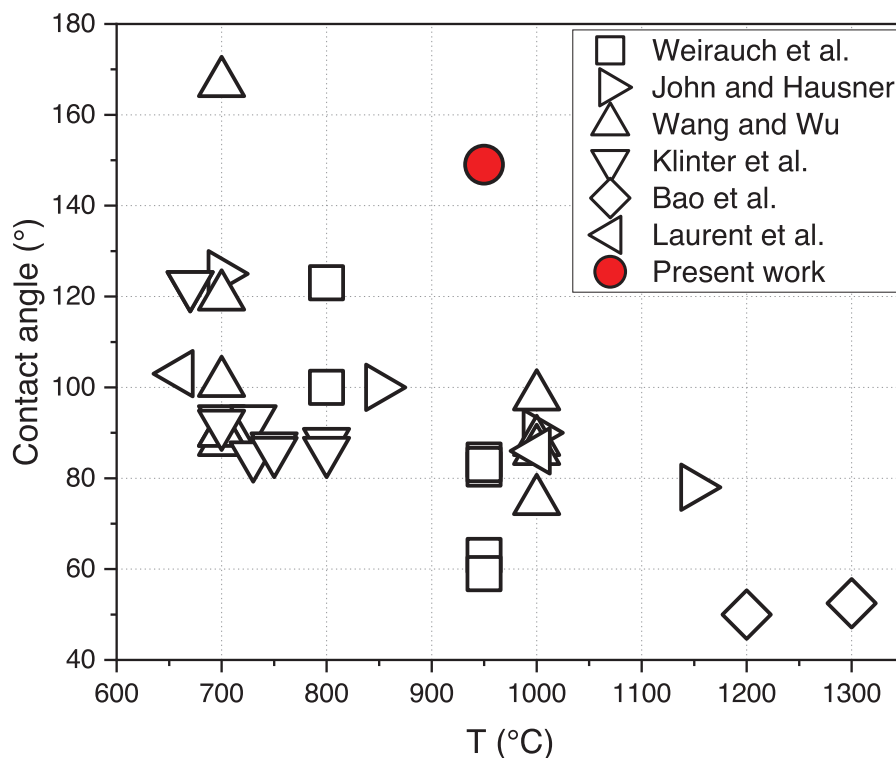


FIGURE 8 The contact angles of molten aluminum on alumina and sapphire substrates.^{24,26,45–48}

TABLE 3 Contact angles of Al and Al-based alloys on mullite and SiO₂ substrates. CP method studies are highlighted; otherwise, it was CH method.

	Temperature (°C)	Roughness (Ra)	Mass	Time (min)	Metal/substrate type	Contact angle θ (°)
Sobczak et al. ⁵⁷	800	–	–	120	Al/mullite	130
	900					94
	1000					52
Voigt et al. ⁴⁴ (CP method)	730	15.8 μm	60 $\pm$ 2 mg	θ showed the same value up to 30 min	AlSi7Mg/mullite	138
Fankhänel et al. ³¹	950	4 μm	60 $\pm$ 2 mg	0 ^a	AlSi7Mg/mullite	132
				120		118
Present work	950	3.1 μm	100 mg	0	Al/mullite	155
Saiz and Tomsia ⁵¹	950	–	0.2–2 g	0	Al/mullite	140
Sobczak et al. (CP method) ⁵⁵	800	–	–	120	Al/SiO ₂	112
Sobczak et al. ⁵⁵						57
Shen et al. ⁴² (CP method)	900	<40 Å	160 mg	0		120
	1100					102
	1250					86
Laurent et al. ⁵⁸	800	15–35 nm	–			150
				120		110
Marumo and Pask ⁵⁶	900	–	100 mg	0		124
Present work	950	12.88 μm	100 mg	0		124

^aThe time at the measuring temperature when the Al sample fully turned to a liquid droplet.

TABLE 4 Contact angle of Al and Al-based alloys on MgAl_2O_4 and MgO substrates.

	Temperature (°C)	Roughness	Mass	Time (min)	Metal/substrate type	Contact angle θ (°)
Shinozaki et al. ⁵⁹	750	Polishing with emery paper (no. 1500)	–	0	Al/ Al_2O_3 -10 mass pct MgO	150
	850–950					140
	1000					125
	750–1000 (100 for each step)			0–60	Al/ Al_2O_3 -28 mass pct MgO	150
Voigt et al. ⁴⁴ (CP method)	730	2.3 μm	60 $\pm$ 2 mg	0	AlSi7Mg/ MgAl_2O_4	110
				0–30		θ fluctuated in range of 110–115
Fankhänel et al. ³¹	950	0.9 μm	60 $\pm$ 2 mg	0	AlSi7Mg/ MgAl_2O_4	151
				120		122
Yoshimi et al. ⁶³	900	–	–	0	Al/ MgO	131
Fujii and Nakae ⁶¹	1100	–	–	θ showed the same value up to 30 min	Al/ MgO	110
Schramm et al. ⁶² (CP method)	680	5.32 μm	60 $\pm$ 2 mg	θ showed a slightly fluctuated value up to 10 min	AlSi7Mg/ MgO	143
Present work	950	3.1 μm	100 mg	0	Al/ MgAl_2O_4	155

Note: CP method studies are highlighted; otherwise, it was CH method.

Voigt et al.⁴⁴ found the lowest contact angle with AlSi7Mg alloy at the lowest temperature point and it slightly fluctuated in the range of 110°–115°, which is interesting. Normally, the lower the temperature, the higher the contact angle. Furthermore, the MgAl_2O_4 substrate of Voigt et al. has a very close roughness value to the present substrate and the investigation of Voigt et al. was carried out via the capillary purification technique. The description of this setup is described in reference.⁶⁰ This setup also allows metals such as magnesium to be tested. The good experimental conditions ensure the absence of an oxide layer on the molten sample, which provides more accurate values. The different measurement techniques can be seen as the reason for this difference in value.

The interaction studies between molten Al and its alloys with MgAl_2O_4 are limited in the literature. A series of interaction studies between molten Al and MgO have been collected as shown in Table 4. Fujii and Nakai⁶¹ divided the contact angle value with time dependence into three stages: 1) initial stage (0–10 s); 2) constant stage; and 3) decreasing stage, based on the obtained results. The authors found that at 900°C, the contact angle remained constant until the end of the interaction experiment. On the other hand, at a higher temperature of 1000°C, the

three stages start to appear. Moreover, when the temperature was increased to 1060°C, the decreasing stage started earlier.⁶¹ In the study of Schramm et al.,⁶² the molten AlSi7Mg alloy showed a high contact angle of 143° on MgO substrate at 680°C. This value was constant up to 10 min interaction time.

3.3 | The interfacial morphology and cross-sectional analysis

The interfacial cross-section of pure Al and Al_2O_3 substrate is presented in Figure 9; no reaction layer was formed. However, one alumina particle was detected at the interface. The detected alumina particle was considered to be formed by the precipitation of oxidized aluminum by dissolved oxygen in the molten aluminum from the side of the alumina substrate. The non-reactive system between molten AlSi7Mg and alumina at 750°C is also stated by Voigt et al.⁴⁴ But the study of ref. [31] reported a reduced contact angle with longer interacting time. The reaction between molten Al and alumina contributes to the contact angle reduction at a higher temperature ($T \geq 950^\circ\text{C}$). In the present study, the interface between Al and alumina

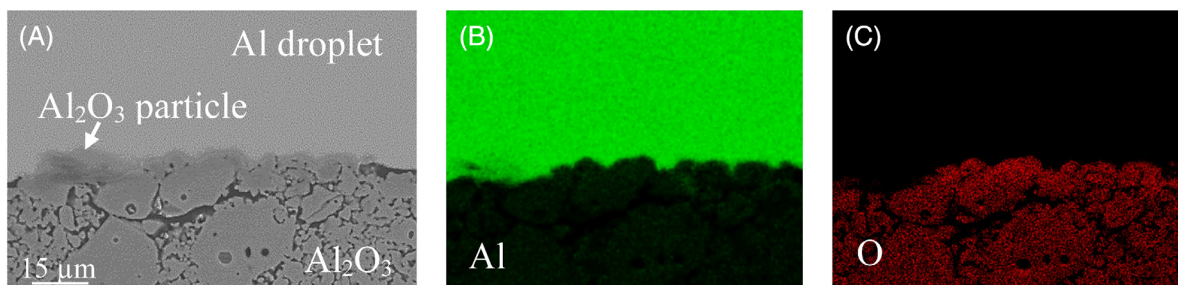


FIGURE 9 (A) Morphology of the cross-section of the solidified Al droplet on Al_2O_3 ceramic substrates. (B) and (C) The elemental mapping of Al and O, respectively.

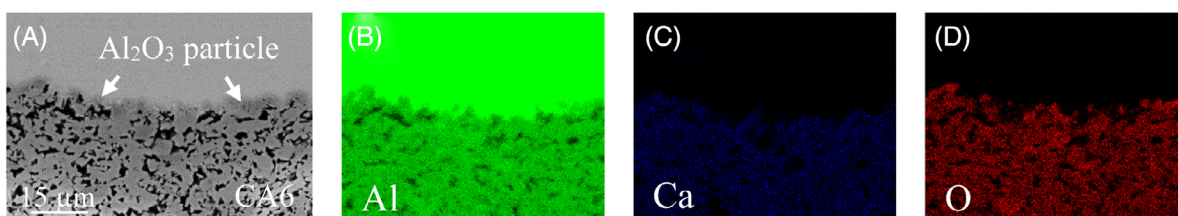


FIGURE 10 (A) The morphology of the solidified Al droplet on CA6 ceramic substrates. (B–D) The elemental mapping of Al, Ca, and O, respectively.

was not significantly modified after the interaction. However, the calculation stated a gas formation consisting of 95.12 mol% Al and 4.88 mol% Al_2O at 10^{-3} Pa. This calculation explains the reduction of the Al_2O_3 to Al_2O (see Equation (1)) and thus the removal of the oxide layer and the reduced contact angle with longer interaction time.

The interaction between molten Al and calcium aluminates (both CA6 and CA2) was not previously studied. In comparison with the other substrates, CA6 substrate showed a less corroded interface, as shown in Figure 10. Nevertheless, a thin layer of aluminum oxide was also detected at the interface. Furthermore, a significant alumina layer at the interface between Al and CA2 was observed, as shown in Figure 10. The thermodynamic calculation for CA6 and Al showed that a small amount of Ca is dissolved in liquid (2×10^{-4} at. fraction) and a small amount of Al_2O_3 is formed, the calculated result is consistent with the obtained experimental result. Moreover, the thermodynamic calculation for CA2 and Al showed that a small amount of Ca is dissolved (with a 1×10^{-3} at. fraction) and a small amount of CA6 is formed. As shown in Figure 11C, an area with high Ca depletion beneath the droplet implies the dissolution of CaO into molten aluminum. Still, the predicted CA6 formation was not observed. It should be remarked that in the cross-section presented in Figure 11A, the CA2 substrate is very dense, only some small pores are scattered.

The interaction between molten Al and mullite was studied by Saiz and Tomsia,⁵¹ who investigated the contact angle as well as the thickness of the reaction layer that formed between molten Al and mullite of different densities at different temperatures (950–1400°C) under purified Ar gas. To compare with the present contact angle result, the density of the ceramic substrates $\geq 95\%$ with the porosity near to 0% was taken into account. The substrates used by Saiz and Tomsia were prepared under varying conditions, as detailed in reference.⁵¹ At 950°C, the present contact angle result (155°) is 15° higher than that determined by Saiz and Tomsia.⁵¹ Fukami et al.⁶⁴ measured the contact angle of liquid iron on MgAl_2O_4 with various porosities (2–39%) and found that the contact angle of iron with the highest porosity of 39% was the highest (130°). Furthermore, the contact angle on the substrate with 27% porosity (approx. 125°), showed an approximately 15° higher value than on the 2% porosity (ca. 110°), a difference that is very close to the present case. Therefore, it is reasonable to conclude that the present higher value is contributed by the higher porosity.

The aluminum contact area after the sessile drop experiments showed an apparent color change visible to the naked eye from light to a dark color. According to the element mapping shown in Figure 12, the SiO_2 in the mullite is reduced to metallic silicon by the liquid aluminum, resulting in a Si depleted zone immediately

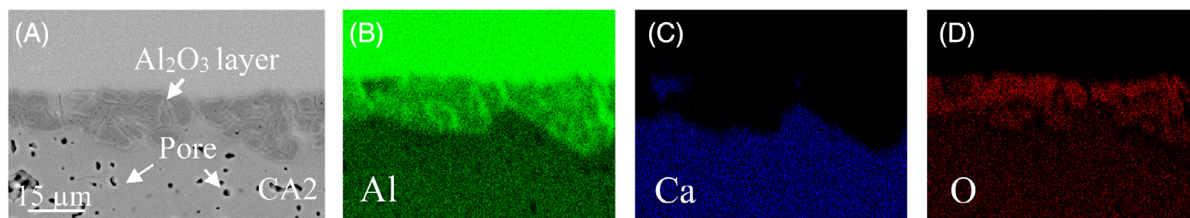


FIGURE 11 (A) The morphology of the solidified Al droplet on CA2 ceramic substrates. (B–D) The elemental mapping of Al, Ca, and O, respectively.

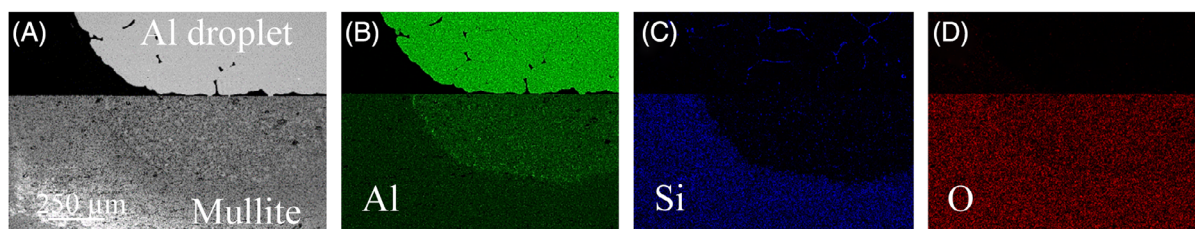


FIGURE 12 (A) The morphology of the solidified Al droplet on CA6 ceramic substrates. (B–D) The elemental mapping of Al, Si, and O, respectively.

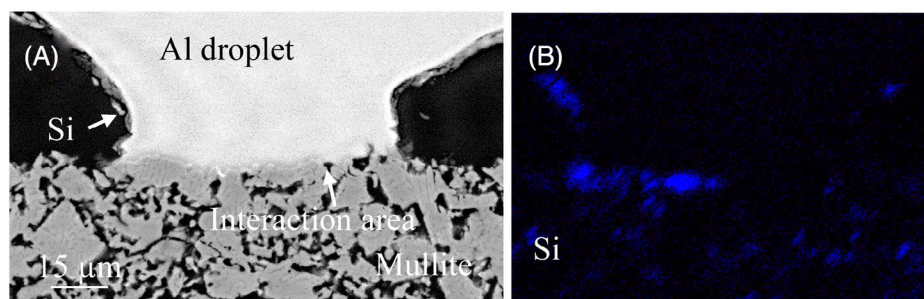
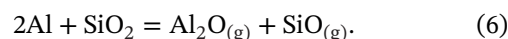
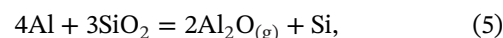
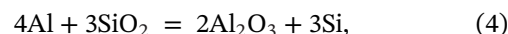


FIGURE 13 (A) Morphology of the cross-section of the solidified Al droplet on mullite ceramic substrates. (B) The elemental mapping of Si.

below the aluminum droplet, as shown in Figure 12B. Besides, the loss of Si leads to the formation of voids in the silica substrate. These voids have the potential to regulate the corrosion degree of molten Al via the increased contact area between molten Al and the silica substrate.⁶⁵ The analogous Si transferring phenomenon has also been reported by Becker et al.³² and Voigt et al.⁴⁴ within the Al–Si alloy and mullite interaction at 950°C. Later, Si was found to precipitate at the interface of pores and in the bulk of the solidified aluminum droplet, as shown in Figure 13. Sobczak⁵⁵ stated that the solid Si precipitates that transferred from the side of SiO₂, mullite, and kaolin materials into molten aluminum alloys tend to nucleate on the Al₂O₃ substrate and its microcrystal and finally resulting in a Si + Al₂O₃ complex. The reducing reaction between molten Al and mullite can be described by Equation (4). The calculated results demonstrated an evidently content of metallic Si (via possible reaction Equations (4 and 5),

and a decreasing tendency with lower pressure. At 0.01 Pa, the gas composition exhibited almost the 99.99 mol% SiO (formed as Equation (6)), 0.006 mol% Al and 0.004 mol% Si.



A similar but more pronounced Al penetration was found in the case of the SiO₂ ceramic substrates, where a massive amount of Si is absorbed by the aluminum, as shown in Figure 14. After the solidification of the aluminum droplet, it was found that the Al-droplet was enriched with Si. Furthermore, the marked spots “1” and “2” in Figure 14A, according to local quantitative

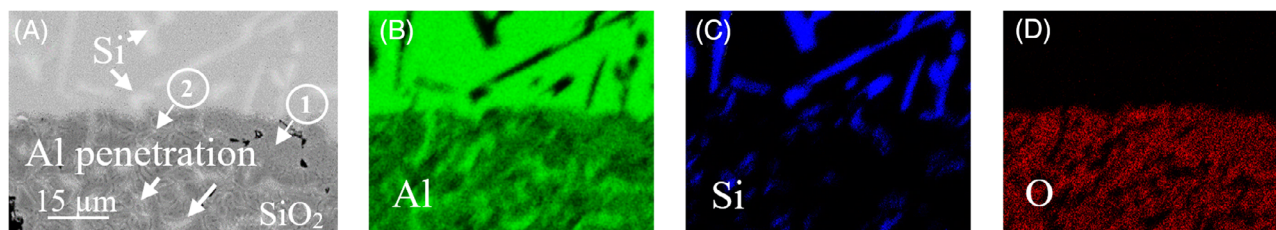


FIGURE 14 (A) The morphology of the solidified Al droplet on SiO_2 ceramic substrates. (B–D) The elemental mapping of Al, Si, and O, respectively.

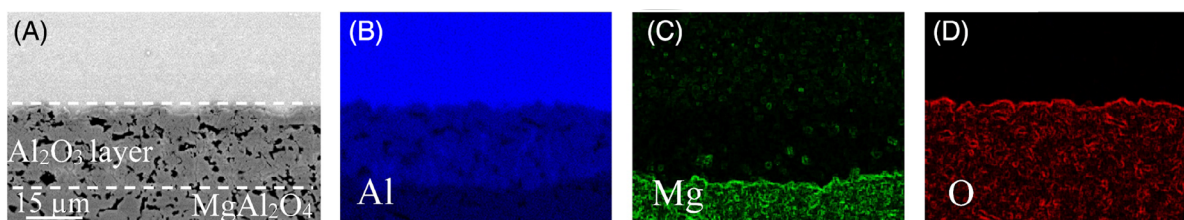


FIGURE 15 (A) The morphology of the solidified Al droplet on MgAl_2O_4 ceramic substrates. (B–D) The elemental mapping of Al, Mg, and O, respectively.

energy-dispersive X-ray spectroscopy, exhibited the composition of Al 68.78 at%, Si 17.24 at%, O 13.98 at%, and Al 57.76 at%, O 41.18 at%, and Si 1.05 at%, respectively. The composition of spot “1” gave an evident hit of SiO_2 reduction. Spot “2” indicates the penetrated aluminum. It was found to contain a rich content of Si. The SiO_2 reduction process is also considered to be mainly controlled by Equation (4). According to the study of Yoshikawa et al.,⁵² the penetration rate and reaction layer thickness are mainly controlled by the Si diffusion rate. In addition, the diffused elementary Si in molten aluminum has the potential to affect the wetting level. Mao et al.⁶⁶ brought molten aluminum containing molten Si (0, 6, 10, 20 mass%) into contact with a graphite substrate at 1000°C. As a result, Si increased the wettability of molten aluminum. In analogous, the transportation of Si during the interaction process reduces the contact angle.

At the MgAl_2O_4 and Al interface, elemental Mg was strongly depleted and remained as an Al_2O_3 layer (see Figure 15). This confirms the investigations by Fankhänel et al.³¹ Furthermore, a slight Al penetration was observed (see Figure 15B). The thermodynamic calculation for MgAl_2O_4 and Al showed that MgAl_2O_4 is transformed into Mg and Al_2O_3 gas, and the solid Al_2O_3 phase shows a constant value, and the spinel phase vanished higher than 14 Pa. This prediction corresponds to the experimentally detected result. Except for that, a small amount of Mg dissolution in liquid Al alloy has also been observed.

4 | CONCLUSIONS

In the present work, the contact angles of molten aluminum on different oxide ceramic substrates (Al_2O_3 , $\text{CaO} \cdot 2\text{Al}_2\text{O}_3$ (CA2), $\text{CaO} \cdot 6\text{Al}_2\text{O}_3$ (CA6), mullite, and SiO_2) were measured in a vacuum heating microscope using the classical sessile drop method at 950°C. The measured contact angles followed the tendency mullite (155°) > CA6 (152°) > MgAl_2O_4 (151°) > Al_2O_3 (149°) > SiO_2 (124°) > CA2 (100°). The adjusted values of the contact angles, taking into account the surface roughness of the substrates according to Wenzel's equation, showed a lower value as follows: CA6 (137.9°) > mullite (132.7°) > Al_2O_3 (131.6°) > MgAl_2O_4 (126.3°) > SiO_2 (100.7°) > CA2 (99.3°). Hence, according to the proposed theory in the previous studies,^{11,21} CA6 and SiO_2 oxides might have a positive effect in stabilizing the aluminum foam. However, the reaction between SiO_2 and molten Al could lead to an enrichment of Si in Al. On the other hand, CA6 exhibited a nonreactive nature toward molten Al. Consequently, CA6 oxide has the potential to be an excellent stabilizer for the Al foam production.

The areas of interaction between the aluminum and the substrates were analyzed. No reaction between Al and Al_2O_3 was observed or predicted. A few Al_2O_3 particles were detected at the interface. This was considered to be precipitation of oxidized aluminum by dissolved oxygen in the molten aluminum from the side of the alumina substrate. SiO_2 in mullite was found to be reduced to Si, and then transferred in the molten aluminum. Solid Si was

then precipitated and, according to the simulation, was in equilibrium with Al_2O_3 . Molten aluminum penetrated the SiO_2 substrate and formed a new Al_2O_3 layer after reducing SiO_2 at the interface. A massive Al_2O_3 layer was formed at the interface during the interaction between molten aluminum and Ca_2 , whereas elemental Ca was strongly depleted. A similar elemental depletion was observed for the MgAl_2O_4 substrate.

ACKNOWLEDGMENTS

The investigation was supported by German Research Foundation (DFG) through the Collaborative Research Center 920 “Multifunctional Filters for Metal Melt Filtration—A contribution towards Zero Defect Materials,” subprojects A03, C01, and C06 at the Technical University Bergakademie Freiberg. X. Wei is very grateful for this opportunity to work with all the researchers involved in this work.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

ORCID

Xingwen Wei  <https://orcid.org/0000-0002-8001-2890>

Mengke Liu  <https://orcid.org/0000-0002-2635-411X>

REFERENCES

- Lefebvre L-P, Banhart J, Dunand DC. Porous metals and metallic foams: current status and recent developments. *Adv Eng Mater.* 2008;10(9):775–87. <https://doi.org/10.1002/adem.200800241>
- Mahadev M, Sreenivasa CG, Shivakumar KM. A review on production of aluminium metal foams. *IOP Conf Ser Mater Sci Eng.* 2018;376:012081. <https://doi.org/10.1088/1757-899X/376/1/012081>
- Banhart J. Metal foams: production and stability. *Adv Eng Mater.* 2006;8(9):781–94. <https://doi.org/10.1002/adem.200600071>
- Banhart J. Manufacturing routes for metallic foams. *JOM.* 2000;52(12):22–27. <https://doi.org/10.1007/s11837-000-0062-8>
- Zhao W, He S, Zhang Y, Zhang C, Tang G, Dai G. Preparation of tin-zinc alloy foam. *Mater Lett.* 2020;266:127482. <https://doi.org/10.1016/j.matlet.2020.127482>
- Wei X, Zhao W, Ilatovskaia M, Perminov A, Liu M, He S, et al. Effect of calcium addition on the aluminum alloy A201 foam. *Adv Eng Mater.* 2024;26(10):2302093. <https://doi.org/10.1002/adem.202302093>
- Zhao W, He S, Wei X, Du Y, Tang G, Zhang Y, et al. Quantitative relationships between cellular structure parameters and the elastic modulus of aluminum foam. *Mater Sci Eng A.* 2023;868:144713. <https://doi.org/10.1016/j.msea.2023.144713>
- Liu XN, Li YX, Chen X, Liu Y, Fan XL. Foam stability in gas injection foaming process. *J Mater Sci.* 2010;45(23):6481–93. <https://doi.org/10.1007/s10853-010-4736-5>
- Wang N, Maire E, Chen X, Adrien J, Li Y, Amani Y, et al. Compressive performance and deformation mechanism of the dynamic gas injection aluminum foams. *Mater Charact.* 2019;147:11–20. <https://doi.org/10.1016/j.matchar.2018.10.013>
- Heim K, Vinod-Kumar GS, García-Moreno F, Banhart J. Stability of various particle-stabilised aluminium alloys foams made by gas injection. *J Mater Sci.* 2017;52(11):6401–14. <https://doi.org/10.1007/s10853-017-0874-3>
- Ip SW, Wang SW, Toguri JM. Aluminum foam stabilization by solid particles. *Can Metall Q.* 1999;38(1):81–92. <https://doi.org/10.1179/cm.1999.38.1.81>
- Wübben T, Odenbach S. Stabilisation of liquid metallic foams by solid particles. *Colloids Surf Physicochem Eng Asp.* 2005;266(1–3):207–13. <https://doi.org/10.1016/j.colsurfa.2005.06.040>
- Yehorov A, Wei X, Bellé MR, Volkova O. Influence of SiO_2 -adding on the thermophysical properties and crystallization behavior of ladle slags. *Steel Res Int.* 2023;2300173. <https://doi.org/10.1002/srin.202300173>
- Liu Z, Pandelaers L, Blanpain B, Guo M. Viscosity of heterogeneous silicate melts: assessment of the measured data and modeling. *ISIJ Int.* 2017;57(11):1895–901. <https://doi.org/10.2355/isijinternational.ISIJINT-2017-288>
- Kadoi K, Nakae H. Relationship between foam stabilization and physical properties of particles on aluminum foam production. *Mater Trans.* 2011;52(10):1912–19. <https://doi.org/10.2320/matertrans.F-M2011817>
- Kaptay G. Interfacial criteria for stabilization of liquid foams by solid particles. *Colloids Surf Physicochem Eng Asp.* 2003;230(1–3):67–80. <https://doi.org/10.1016/j.colsurfa.2003.09.016>
- Kumar GSV, Chakraborty M, García-Moreno F, Banhart J. Foamability of MgAl_2O_4 (spinel)-reinforced aluminum alloy composites. *Metall Mater Trans A.* 2011;42(9):2898–908. <https://doi.org/10.1007/s11661-011-0709-9>
- Kumar GSV, Heim K, Mukherjee M, García-Moreno F, Banhart J. Investigation on the stabilization behaviour of MgAl_2O_4 (spinel) particles in aluminium foam via in-situ X-ray radiography and FIB tomography. *Mater Today Commun.* 2024;38:108257. <https://doi.org/10.1016/j.mtcomm.2024.108257>
- Bhosale D, Georgy K, Mukherjee M, Kumar GSV. Production, stability and properties of ultrafine MgAl_2O_4 (spinel) particles stabilized Mg–3Ca alloy foams. *J Mater Res Technol.* 2024;28:4012–24. <https://doi.org/10.1016/j.jmrt.2024.01.039>
- Heim K, García-Moreno F, Banhart J. Particle size and fraction required to stabilise aluminium alloy foams created by gas injection. *Scr Mater.* 2018;153:54–58. <https://doi.org/10.1016/j.scriptamat.2018.04.041>
- Haibel A, Rack A, Banhart J. Why are metal foams stable? *Appl Phys Lett.* 2006;89(15):154102. <https://doi.org/10.1063/1.2357931>
- Ip SW, Kucharski M, Toguri JM. Wetting behaviour of aluminium and aluminium alloys on Al_2O_3 and CaO . *J Mater Sci Lett.* 1993;12:1699–702. <https://doi.org/10.1007/BF00418836>
- Humenik Jr. M, Kingery WD. Metal-ceramic interactions: III, surface tension and wettability of metal-ceramic systems. *J Am Ceram Soc.* 1954;37(1):18–23. <https://doi.org/10.1111/j.1151-2916.1954.tb13972.x>
- Bao S, Tang K, Kvithyld A, Tangstad M, Engh TA. Wettability of aluminum on alumina. *Metall Mater Trans B.* 2011;42(6):1358–66. <https://doi.org/10.1007/s11663-011-9544-z>
- Sobczak N, Singh M, Asthana R. High-temperature wettability measurements in metal/ceramic systems—some methodological issues. *Curr Opin Solid State Mater Sci.*

- 2005;9(4–5):241–53. <https://doi.org/10.1016/j.cossms.2006.07.007>
26. Weirauch DA, Krafick WJ. The effect of carbon on wetting of aluminum oxide by aluminum. *Metall Trans A*. 1990;21(6):1745–51. <https://doi.org/10.1007/BF02672591>
27. Yuan J, Li Y, Zhou Y. Effect of contact angle on bubble formation at submerged orifices. *J Mater Sci*. 2014;49(23):8084–94. <https://doi.org/10.1007/s10853-014-8516-5>
28. Wei X, Dudczig S, Storti E, Ilatovskaia M, Endo R, Aneziris CG, et al. Interaction of molten Armco iron with various ceramic substrates at 1600°C. *J Eur Ceram Soc*. 2022;42:2535–44. <https://doi.org/10.1016/j.jeurceramsoc.2022.01.011>
29. Wei X, Yehorov A, Storti E, Dudczig S, Fabrichnaya O, Aneziris CG, et al. Phenomenon of whiskers formation in Al_2O_3 –C refractories. *Adv Eng Mater*. 2021;24(2):2100718. <https://doi.org/10.1002/adem.202100718>
30. Wei X, Storti E, Dudczig S, Fabrichnaya O, Aneziris CG, Volkova O. Interactions between molten iron and carbon bonded filter materials. In: Aneziris CG, Biermann H, editors. Multifunctional ceramic filter systems for metal melt filtration Vol. 337. Cham: Springer International Publishing; 2024. p. 533–50. https://doi.org/10.1007/978-3-031-40930-1_21
31. Fankhänel B, Stelter M, Voigt C, Aneziris CG. Interaction of AlSi7Mg with oxide ceramics: interaction of AlSi7Mg with oxide. *Adv Eng Mater*. 2017;19(9):1700084. <https://doi.org/10.1002/adem.201700084>
32. Becker H, Fankhänel B, Voigt C, Charitos A, Stelter M, Aneziris CG, et al. Interaction of Fe-containing, secondary Al–Si alloy with oxide and carbon-containing ceramics for Fe removal. *Adv Eng Mater*. 2021;24(2):2100595. <https://doi.org/10.1002/adem.202100595>
33. Zienert T, Fabrichnaya O. Thermodynamic assessment and experiments in the system $\text{MgO-Al}_2\text{O}_3$. *Calphad*. 2013;40:1–9. <https://doi.org/10.1016/j.calphad.2012.10.001>
34. Liang P, Su H-L, Seifert HJ, et al. Experimental investigation and thermodynamic calculation of the central part of the Mg–Al phase diagram. *Z Fuer Met*. 1998;89:536–40.
35. Hallstedt B. Thermodynamic calculation of some subsystems of the Al–Ca–Mg–Si–O system. *J Phase Equilibria*. 1993;14(6):662–75. <https://doi.org/10.1007/BF02667878>
36. Hallstedt B. The magnesium–oxygen system. *Calphad*. 1993;17(3):281–86. [https://doi.org/10.1016/0364-5916\(93\)90006-W](https://doi.org/10.1016/0364-5916(93)90006-W)
37. Hallstedt B. Assessment of the $\text{CaO-Al}_2\text{O}_3$ system. *J Am Ceram Soc*. 1990;73(1):15–23. <https://doi.org/10.1111/j.1151-2916.1990.tb05083.x>
38. Selleby M, Sundman B. A reassessment of the Ca–Fe–O system. *Calphad*. 1996;20(3):381–92. [https://doi.org/10.1016/S0364-5916\(96\)00039-9](https://doi.org/10.1016/S0364-5916(96)00039-9)
39. Dumitrescu L, Sundman B. A thermodynamic reassessment of the Si–Al–O–N system. *J Eur Ceram Soc*. 1995;15(3):239–47. [https://doi.org/10.1016/0955-2219\(95\)93945-Y](https://doi.org/10.1016/0955-2219(95)93945-Y)
40. Feufel H, Gödecke T, Lukas HL, Sommer F. Investigation of the Al–Mg–Si system by experiments and thermodynamic calculations. *J Alloys Compd*. 1997;247(1):31–42. [https://doi.org/10.1016/S0925-8388\(96\)02655-2](https://doi.org/10.1016/S0925-8388(96)02655-2)
41. Dinsdale AT. SGTE Substance Database—SGTE—Scientific Group Thermodata Europe. *CALPHAD*. 1991;15(4):317–425.
42. Shen P, Fujii H, Matsumoto T, Nogi K. Reactive wetting of SiO_2 substrates by molten Al. *Metall Mater Trans A*. 2004;35(2):583–88. <https://doi.org/10.1007/s11661-004-0369-0>
43. Gu Y, Zhang W, Mou J, Zheng S, Jiang L, Sun Z, et al. Research progress of biomimetic superhydrophobic surface characteristics, fabrication, and application. *Adv Mech Eng*. 2017;9(12):168781401774685. <https://doi.org/10.1177/1687814017746859>
44. Voigt C, Ditscherlein L, Werzner E, Zienert T, Nowak R, Peuker U, et al. Wettability of AlSi7Mg alloy on alumina, spinel, mulite and rutile and its influence on the aluminum melt filtration efficiency. *Mater Des*. 2018;150:75–85. <https://doi.org/10.1016/j.matdes.2018.04.026>
45. John H, Hausner H. Wetting of aluminum oxide by liquid aluminum. *Int J High Technol Ceram*. 1986;2(1):73–78. [https://doi.org/10.1016/0267-3762\(86\)90006-8](https://doi.org/10.1016/0267-3762(86)90006-8)
46. Wang D-J, Wu S-T. The influence of oxidation on the wettability of aluminum on sapphire. *Acta Metall Mater*. 1994;42(12):4029–34. [https://doi.org/10.1016/0956-7151\(94\)90180-5](https://doi.org/10.1016/0956-7151(94)90180-5)
47. Klintner AJ, Mendoza-Suarez G, Drew RAL. Wetting of pure aluminum and selected alloys on polycrystalline alumina and sapphire. *Mater Sci Eng A*. 2008;495(1–2):147–52. <https://doi.org/10.1016/j.msea.2007.10.113>
48. Laurent V, Chatain D, Chatillon C, Eustathopoulos N. Wettability of monocrystalline alumina by aluminium between its melting point and 1273 K. *Acta Metall*. 1988;36(7):1797–803. [https://doi.org/10.1016/0001-6160\(88\)90248-9](https://doi.org/10.1016/0001-6160(88)90248-9)
49. Jacob, KT, Srikanth, S. Physical chemistry of the reduction of calcium oxide with aluminum in vacuum. *High Temp Mater Process*. 1990;9(2–4):77–92. <https://doi.org/10.1515/HTMP.1990.9.2-4.77>
50. Prabritputaloong K, Piggott MR. Thin-film studies of the reduction of SiO_2 by Al. *J Am Ceram Soc*. 1973;56(4):177–80. <https://doi.org/10.1111/j.1151-2916.1973.tb12449.x>
51. Saiz E, Tomsia AP. Kinetics of metal–ceramic composite formation by reactive penetration of silicates with molten aluminum. *J Am Ceram Soc*. 1998;81(9):2381–93. <https://doi.org/10.1111/j.1151-2916.1998.tb02634.x>
52. Yoshikawa N, Hattori A, Taniguchi S. Growth rate of reaction layer between SiO_2 and molten Al above 1473 K. *Mater Trans*. 2005;46(4):842–45. <https://doi.org/10.2320/matertrans.46.842>
53. Standage AE, Gani MS. Reaction between vitreous silica and molten aluminum. *J Am Ceram Soc*. 1967;50(2):101–5. <https://doi.org/10.1111/j.1151-2916.1967.tb15049.x>
54. Shen P, Fujii H, Matsumoto T, Nogi K. Critical factors affecting the wettability of α -alumina by molten aluminum. *J Am Ceram Soc*. 2004;87(7):1265–73. <https://doi.org/10.1111/j.1151-2916.2004.tb07721.x>
55. Sobczak N. Interaction between molten aluminum and oxides In: Gupta N, Hunt WH, editors. Solidification Processing of MetalMatrix Composites—Rohatgi Honorary Symposium (TMS, TheMinerals, Metals & Materials Society, 2006), pp. 133–46.
56. Marumo C, Pask JA. Reactions and wetting behaviour in the aluminium–fused silica system. *J Mater Sci*. 1977;12(2):223–33. <https://doi.org/10.1007/BF00566262>
57. Sobczak N. Wettability and reactivity between molten aluminum and selected oxides. *Solid State Phenom*. 2005;

- 101–102:221–26. <https://doi.org/10.4028/www.scientific.net/SSP.101-102.221>
58. Laurent V, Chatain D, Eustathopoulos N. Wettability of SiO_2 and oxidized SiC by aluminium. *Mater Sci Eng A*. 1991;135:89–94. [https://doi.org/10.1016/0921-5093\(91\)90542-U](https://doi.org/10.1016/0921-5093(91)90542-U)
59. Shinozaki N, Mukai K, Fujita T. Wettability of Al_2O_3 -MgO substrates by molten aluminum. *Metall Mater Trans B*. 2002;33(3):506–9. <https://doi.org/10.1007/s11663-002-0062-x>
60. Kudyba A, Sobczak N, Budzioch J, Polkowski W, Giuranno D. Improvements in experimental investigation of molten Mg-based materials. *Mater Des*. 2018;160:915–17. <https://doi.org/10.1016/j.matdes.2018.10.028>
61. Fujii H, Nakae H. Equilibrium contact angle in the magnesium oxide/aluminium system. *Acta Mater*. 1996;44(9):3567–73. [https://doi.org/10.1016/1359-6454\(96\)00017-1](https://doi.org/10.1016/1359-6454(96)00017-1)
62. Schramm A, Nowak R, Bruzda G, Polkowski W, Fabrichnaya O, Aneziris CG. High temperature wettability and corrosion of ZrO_2 , Al_2O_3 , Al_2O_3 -C, MgO and MgAlON ceramic substrates by an AZ91 magnesium alloy melt. *J Eur Ceram Soc*. 2022;42(6):3023–35. <https://doi.org/10.1016/j.jeurceramsoc.2022.01.040>
63. Yoshimi N, Nakae H, Fujii H. A new approach to estimating wetting in a reaction system. *Mater Trans JIM*. 1990;31(2):141–47. <https://doi.org/10.2320/matertrans1989.31.141>
64. Fukami N, Wakamatsu R, Shinozaki N, Wasai K. Wettability between porous MgAl_2O_4 substrates and molten Iron. *Mater Trans*. 2009;50(11):2552–56. <https://doi.org/10.2320/matertrans.M2009204>
65. Moya JS, Steier HP, Requena J. Interfacial reactions in aluminum alloys/mullite–zirconia composites. *Compos Part Appl Sci Manuf*. 1999;30(4):439–44. [https://doi.org/10.1016/S1359-835X\(98\)00132-8](https://doi.org/10.1016/S1359-835X(98)00132-8)
66. Mao W, Noji T, Teshima K, Shinozaki N. Wettability of molten aluminum-silicon alloys on graphite and surface tension of those alloys at 1273 K (1000°C). *Metall Mater Trans A*. 2016;47(6):3201–12. <https://doi.org/10.1007/s11661-016-3460-4>

How to cite this article: Wei X, Fankhänel B, Zhao W, Perminov A, Storti E, Liu M, et al. The wettability of pure liquid aluminum on oxide substrates. *J Am Ceram Soc*. 2025;108:e20103. <https://doi.org/10.1111/jace.20103>